ESGEM-AMR

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AMRrules

amrrules.org

Agenda

1. AMRrules Python package - for March release

- Genome summary reports
- Finalising core gene rules

2. AMRrulebrowser

3. AMRgen R package

- New functions

4. Planning

- Geno-pheno data-driven rules
- Manuscript plans

https://github.com/AMRverse

AMRrules Public
Forked from katholt/AMRrules

Interpretative standards for AMR genotypes

Python 27 4

ESGEM-AMR Public

This is the webpage of the ESGEM-AMR Working Group

HTML 15 4

AMRgen Public

The AMRgen package provides tools for interpreting antimicrobial resistance (AMR) genes, integrating genomic data with phenotypic antimicrobial susceptibility testing (AST) data, and calculating ge...

R 15 9

AMRrulemakeR Public

What the Package Does (One Line, Title Case)

R 1

AMRrulebrowser Public

Web browser to interactively explore AMRrules

JavaScript 1

AMRrulevalidator Public

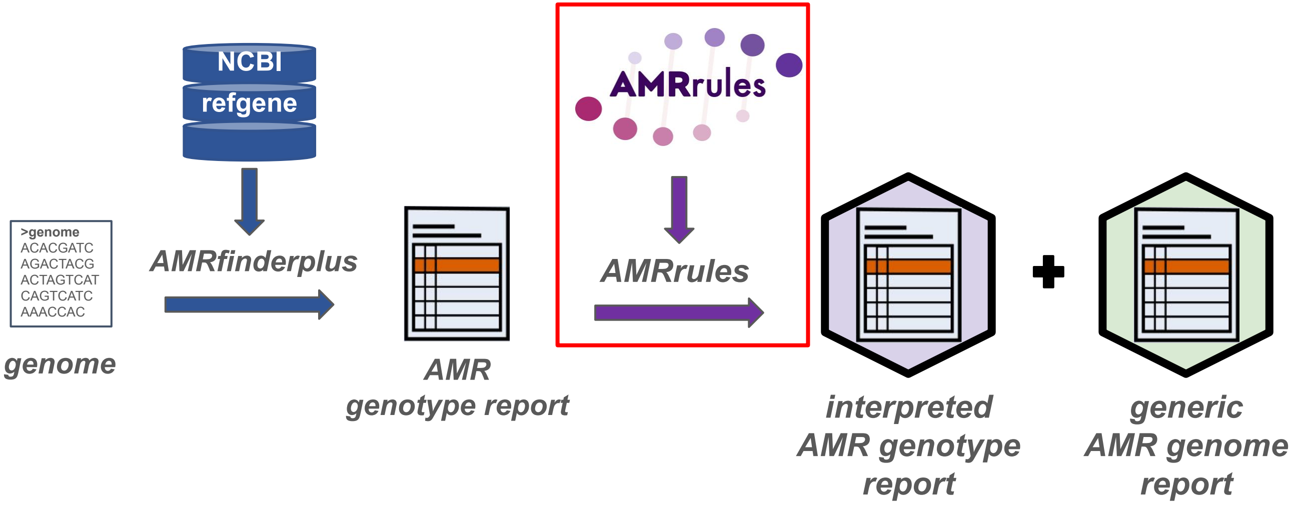
Validates draft AMRrules files before importation into interpretation engine

Python

AMRrules package - written in Python

Development branch: https://github.com/AMRverse/AMRrules/**tree/genome_summary_report_dev**

Package includes
- rules
- software to interpret AMRfinderplus genotype results using the rules



Rules in first beta release:

- Acinetobacter baumannii
- Enterobacter
- Enterococcus faecalis
- Enterococcus faecium
- Escherichia coli
- Klebsiella pneumoniae
- Neisseria gonorrhoeae (acquired resistances,
- Pseudomonas aeruginosa
- Salmonella
- Staphylococcus aureus
- Yersinia

Output 1: Annotated AMRfinderplus report

(v1-in dev)

NCBI AMRfinderplus

AMRrules genotype interpretation

Gene symbol	Subclass	Hierarchy node	Variation type	mutation	ruleID	gene context	drug	drug class	phenotype	clinical category	evidence grade	version	organism
emrD	EFFLUX	emrD	Gene presence detected	-	-	-	-	-	-	-	-	0.1.0	-
oqxB19	PHENICOL/ QUINOLONE	oqxB19	Gene presence detected	-	KPN0003	core	ciprofloxacin	-	wildtype	S	moderate	0.1.0	s__Klebsiella pneumoniae
oqxA	PHENICOL/ QUINOLONE	oqxA	Gene presence detected	-	KPN0002	core	ciprofloxacin	-	wildtype	S	moderate	0.1.0	s__Klebsiella pneumoniae
blaSHV-11	BETA-LACTAM	blaSHV-11	Gene presence detected	-	KPN0001	core	-	penicillin beta-lactam	wildtype	R	high	0.1.0	s__Klebsiella pneumoniae
fosA	FOSFOMYCIN	fosA5_fam	Gene presence detected	-	KPN0004	core	fosfomycin	-	wildtype	S	moderate	0.1.0	s__Klebsiella pneumoniae

- Default: ‘minimal’ annotation (as shown here)
- ‘full’ option additionally contains:
  - breakpoint, breakpoint standard, evidence code, evidence limitations, PMID, rule curation note

Output 2: Genome summary report

Klebsiella pneumoniae SGH10 (ampR)

drug	drug class	clinical category	phenotype	evidence grade	markers (non-S)	markers (no rule)	markers (S)	ruleIDs	combo rules	organism
ciprofloxacin	fluoroquinolone antibiotic	S	wildtype	moderate	-	-	oqxB19;oqxA	KPN0002; KPN0003	-	s__Klebsiella pneumoniae
-	penicillin beta-lactam	R	wildtype	high	blaSHV-11	-	-	KPN0001	-	s__Klebsiella pneumoniae
fosfomycin	phosphonic acid antibiotic	S	wildtype	moderate	-	-	fosA	KPN0004	-	s__Klebsiella pneumoniae
-	antibiotic efflux	-	-	-	-	emrD	-	-	-	s__Klebsiella pneumoniae

markers assigned to columns based on rule

overall interpretation for each drug/class based on markers and rules

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Updates to AMRrules genome summary output

- Now includes separate row for ‘partial’ hits

drug	drug class	clinical category	phenotype	evidence grade	markers (non-S)	markers (no rule)	markers (S)	ruleIDs	combo rules	organism
ciprofloxacin	fluoroquinolone antibiotic	S	wildtype	moderate	-	-	oqxB19;oqxA	KPN0002; KPN0003	-	s__Klebsiella pneumoniae
-	penicillin beta-lactam	R	wildtype	high	blaSHV-11	-	-	KPN0001	-	s__Klebsiella pneumoniae
fosfomycin	phosphonic acid antibiotic	S	wildtype	moderate	-	-	fosA	KPN0004	-	s__Klebsiella pneumoniae
-	antibiotic efflux	-	-	-	-	emrD	-	-	-	s__Klebsiella pneumoniae
-	partial	-	-	-	-	catB3:-; aac(3)-IIe:-	-	-	-	s__Klebsiella pneumoniae

Hits with an AMRFP Method starting with “PARTIAL” are placed here, no interpretation

Updates to AMRrules genome summary output

- Now includes separate row for ‘partial’ hits

- More options for how to handle markers without rules
    - *To discuss:* Which should be default?

How to report markers with no rule?

```
amrrules --no-rule-interpretation nwtR
```

Klebsiella pneumoniae SAMD00055732 (multidrug resistant)

drug	drug class	clinical category	phenotype	evidence grade	markers (non-S)	markers (no rule)	markers (S)	ruleIDs	combo rules	organism
gentamicin	aminoglycoside antibiotic	R	nonwildtype	very low	-	aac(3)-IId;aac(6')-Ib4	-	-	-	s__Klebsiella pneumoniae
streptomycin	aminoglycoside antibiotic	R	nonwildtype	very low	-	aph(3'')-Ib;aph(6)-Id	-	-	-	s__Klebsiella pneumoniae
-	carbapenem	R	nonwildtype	very low	-	blaIMP-4	-	-	-	s__Klebsiella pneumoniae
-	penicillin beta-lactam	R	nonwildtype	high	blaSHV-1	blaTEM-1	-	KPN0001	-	s__Klebsiella pneumoniae
fosfomycin	phosphonic acid antibiotic	S	wildtype	moderate	-	-	fosA	KPN0004	-	s__Klebsiella pneumoniae
-	sulfonamide antibiotic	R	nonwildtype	very low	-	sul2;sul1	-	-	-	s__Klebsiella pneumoniae
-	antibiotic efflux	-	-	-	-	emrD	-	-	-	s__Klebsiella pneumoniae

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How to report markers with no rule?

```
amrrules --no-rule-interpretation nwtS
```

Klebsiella pneumoniae SAMD00055732 (multidrug resistant)

drug	drug class	clinical category	phenotype	evidence grade	markers (non-S)	markers (no rule)	markers (S)	ruleIDs	combo rules	organism
gentamicin	aminoglycoside antibiotic	S	nonwildtype	very low	-	aac(3)-IId;aac(6')-Ib4	-	-	-	s__Klebsiella pneumoniae
streptomycin	aminoglycoside antibiotic	S	nonwildtype	very low	-	aph(3")-Ib; aph(6)-Id	-	-	-	s__Klebsiella pneumoniae
-	carbapenem	S	nonwildtype	very low	-	blaIMP-4	-	-	-	s__Klebsiella pneumoniae
-	penicillin beta-lactam	R	nonwildtype	high	blaSHV-1	blaTEM-1	-	KPN0001	-	s__Klebsiella pneumoniae
fosfomycin	phosphonic acid antibiotic	S	wildtype	moderate	-	-	fosA	KPN0004	-	s__Klebsiella pneumoniae
-	sulfonamide antibiotic	S	nonwildtype	very low	-	sul2;sul1	-	-	-	s__Klebsiella pneumoniae
-	antibiotic efflux	-	-	-	-	emrD	-	-	-	s__Klebsiella pneumoniae

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How to report markers with no rule?

```
amrrules --no-rule-interpretation nwt
```

Klebsiella pneumoniae SAMD00055732 (multidrug resistant)

drug	drug class	clinical category	phenotype	evidence grade	markers (non-S)	markers (no rule)	markers (S)	ruleIDs	combo rules	organism
gentamicin	aminoglycoside antibiotic	-	nonwildtype	very low	-	aac(3)-IId;aac(6')-Ib4	-	-	-	s__Klebsiella pneumoniae
streptomycin	aminoglycoside antibiotic	-	nonwildtype	very low	-	aph(3'')-Ib;aph(6)-Id	-	-	-	s__Klebsiella pneumoniae
-	carbapenem	-	nonwildtype	very low	-	blaIMP-4	-	-	-	s__Klebsiella pneumoniae
-	penicillin beta-lactam	R	nonwildtype	high	blaSHV-1	blaTEM-1	-	KPN0001	-	s__Klebsiella pneumoniae
fosfomycin	phosphonic acid antibiotic	S	wildtype	moderate	-	-	fosA	KPN0004	-	s__Klebsiella pneumoniae
-	sulfonamide antibiotic	-	nonwildtype	very low	-	sul2;sul1	-	-	-	s__Klebsiella pneumoniae
-	antibiotic efflux	-	-	-	-	emrD	-	-	-	s__Klebsiella pneumoniae

How to report markers with no rule?

```
amrrules --no-rule-interpretation none
```

Klebsiella pneumoniae SAMD00055732 (multidrug resistant)

drug	drug class	clinical category	phenotype	evidence grade	markers (non-S)	markers (no rule)	markers (S)	ruleIDs	combo rules	organism
gentamicin	aminoglycoside antibiotic	-	-	very low	-	aac(3)-IId;aac(6')-Ib4	-	-	-	s__Klebsiella pneumoniae
streptomycin	aminoglycoside antibiotic	-	-	very low	-	aph(3")-Ib;aph(6)-Id	-	-	-	s__Klebsiella pneumoniae
-	carbapenem	-	-	very low	-	blaIMP-4	-	-	-	s__Klebsiella pneumoniae
-	penicillin beta-lactam	R	wildtype	high	blaSHV-1	blaTEM-1	-	KPN0001	-	s__Klebsiella pneumoniae
fosfomycin	phosphonic acid antibiotic	S	wildtype	moderate	-	-	fosA	KPN0004	-	s__Klebsiella pneumoniae
-	sulfonamide antibiotic	-	-	very low	-	sul2;sul1	-	-	-	s__Klebsiella pneumoniae
-	antibiotic efflux	-	-	-	-	emrD	-	-	-	s__Klebsiella pneumoniae

Example genome summary report: no rule parameter

What should be the default setting?

`--no-rule-interpretation` options:

- nwtR (current default)
- nwtS
- nwt
- none

Note that 'nwt' and 'none' options still to be implemented, current version of code only supports 'nwtR' and 'nwtS'

Defining rules: updates

1. **Remove condition-specific rules**

- Do not define condition-specific rules
  - If you find an example where you really want to, let us know to discuss

E.g. *E. coli* rules included condition-specific rules

Cephalosporins[1]	MIC breakpoints (mg/L)			Disk content (µg)	Zone diameter breakpoints (mm)		
	S ≤	R >	ATU		S ≥	R <	ATU
Cefuroxime iv, *E. coli*, *Klebsiella* spp. (except *K. aerogenes*), *Raoultella* spp. and *P. mirabilis*	0.001	8		30	50	19	
Cefuroxime oral (uncomplicated UTI only), *E. coli, Klebsiella* spp. (except *K. aerogenes*), *Raoultella* spp. and *P. mirabilis*	8	8		30	19	19	

ruleID	nodeID	variation type	gene context	drug	drug class	phenotype	clinical category	breakpoint condition	
ECO0007	blaEC	Gene presence detected	core	-	second-generation cephalosporin	wildtype	S	-	
ECO0008	blaEC	Gene presence detected	core	cefoxitin	-	wildtype	S	-	
ECO0009	blaEC	Gene presence detected	core	cefuroxime	-	wildtype	S	Oral	
ECO0010	blaEC	Gene presence detected	core	cefuroxime	-	wildtype	I	Intravenous	*removed*

Defining rules: updates

1. **Remove condition-specific rules**
- Do not define condition-specific rules
  - If you find an example where you really want to, let us know to discuss

2. **Define rules for loss of R-associated core genes:**
- If you have core genes associated with a WT R rule -> define explicit rule for disruption of that gene (presumably WT S)

Rules for loss of WT R core genes

Rules for core gene blaSHV

ruleID	txid	organism	gene	nodeID	mutation	variation type	gene context	drug	drug class	phenotype	clinical category	breakpoint
KPN0001	573	s__Klebsiella pneumoniae	blaSHV	blaSHV	-	Gene presence detected	core	-	penicillin beta-lactam	wildtype	R	not applicable
KPN0030	573	s__Klebsiella pneumoniae	blaSHV	blaSHV	-	Inactivating mutation detected	core	-	penicillin beta-lactam	wildtype	S	not applicable

AMRfinderplus output

AMRrules annotated genotype report

Gene symbol	Method	Hierarchy node	variation type	gene	mutation	ruleID
blaSHV-11	EXACTX	blaSHV	Gene presence detected	blaSHV	-	KPN0001
blaSHV-11	PARTIAL_CONTIG_ENDX	blaSHV	Inactivating mutation detected	blaSHV	-	KPN0030

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Core gene detected: WT R

Annotated genotype report

Gene symbol	Subclass	Hierarchy node	Variation type	mutation	ruleID	gene context	drug	drug class	phenotype	clinical category	evidence grade	version	organism
blaSHV-11	BETA-LACTAM	blaSHV-11	Gene presence detected	-	KPN0030	core	-	penicillin beta-lactam	wildtype	R	high	0.1.0	s__Klebsiella pneumoniae

Genome summary

drug	drug class	clinical category	phenotype	evidence grade	markers (non-S)	markers (no rule)	markers (S)	ruleIDs	combo rules	organism
ciprofloxacin	fluoroquinolone antibiotic	S	wildtype	moderate	-	-	oqxB19;oqxA	KPN0002; KPN0003	-	s__Klebsiella pneumoniae
-	penicillin beta-lactam	R	wildtype	high	blaSHV-11	-	-	KPN0001	-	s__Klebsiella pneumoniae
fosfomycin	phosphonic acid antibiotic	S	wildtype	moderate	-	-	fosA	KPN0004	-	s__Klebsiella pneumoniae
-	antibiotic efflux	-	-	-	-	emrD	-	-	-	s__Klebsiella pneumoniae

Core gene inactivated: WT S

Annotated genotype report

Gene symbol	Subclass	Hierarchy node	Variation type	mutation	ruleID	gene context	drug	drug class	phenotype	clinical category	evidence grade	version	organism
blaSHV-11	BETA-LACTAM	blaSHV-11	Inactivating mutation detected	-	KPN0030	core	-	penicillin beta-lactam	wildtype	S	high	0.1.0	s__Klebsiella pneumoniae

Genome summary

drug	drug class	clinical category	phenotype	evidence grade	markers (non-S)	markers (no rule)	markers (S)	ruleIDs	combo rules	organism
ciprofloxacin	fluoroquinolone antibiotic	S	wildtype	moderate	-	-	oqxB19;oqxA	KPN0002; KPN0003	-	s__Klebsiella pneumoniae
-	penicillin beta-lactam	S	wildtype	high	blaSHV-11	-	-	KPN0030	-	s__Klebsiella pneumoniae
fosfomycin	phosphonic acid antibiotic	S	wildtype	moderate	-	-	fosA	KPN0004	-	s__Klebsiella pneumoniae
-	antibiotic efflux	-	-	-	-	emrD	-	-	-	s__Klebsiella pneumoniae

AMRrules package - steps to v1.0 release

1. **Feb 27:** deadline to submit/modify core gene rules for v1 release
2. **Feb 27:** deadline for feedback on software and output formats
3. Early March:
    a. Incorporate feedback on software, default parameters, and output formats
    b. Incorporate additional core gene rules submitted
        - *Bordetella, Shewanella, Burkholderia pseudomallei, Legionella, Proteus mirabilis, Achromobacter xylosoxidans, Neisseria meningitidis, Klebsiella oxytoca, Mycobacterium tuberculosis(?)*

AIM: Release by mid March (ahead of Wellcome AMR Conference, March 23, where Kat will present on AMRrules)

Please provide feedback on the AMRrules interpretation engine

- Look at example outputs in Google sheet:
  - https://docs.google.com/spreadsheets/d/1dvllmkoGSi_ILuzDQ5wTVYL3YWxPQh161Qs3FW4ui1c/edit?usp=sharing
- Look at test outputs in AMRrules repo `tests/` (development branch)
  - https://github.com/AMRverse/AMRrules/tree/genome_summary_report_dev/tests/data/example_output
- Test AMRrules package (development branch) on your own genomes
  - https://amrrules.readthedocs.io/en/genome_summary_report_dev/installation.html

How to share feedback

- Post an issue: https://github.com/AMRverse/AMRrules/issues
- Slack channel #codequeries
- Email esgem.amr@gmail.com

AMRrules Documentation

https://amrrules.readthedocs.io/en/**genome_summary_report_dev/installation.html**

How to test

Search

AMRrules
Installation
Rules and supported organisms
Usage
Interpretation logic and output
AMRrules Specification
Test Data

Download and install AMRrules

Currently, AMRrules is only available for installation via source. We recommend you set up a new conda environment while the tool is under development.:

```
# create your conda environment
conda create -n amrrules_beta -c bioconda python=3.12 pip
conda activate amrrules_beta

# clone the repository, switch to the development branch
git clone https://github.com/AMRverse/AMRrules
cd AMRrules
git checkout genome_summary_report_dev

# install AMRrules
make dev
```

After installation, you must download the required AMRFinderPlus resource files. Run:

```
amrrules --download-resources
```

This will download and cache the necessary files for AMRrules to function. You only need to run this **once** after installation, or when updating resources (eg a new AMRFinderPlus database has been released).

Check which organisms have rule-sets available in the installation:

```
amrrules --list-organisms
```

Test on the included example datasets:

```
amrrules --input tests/data/input/test_ecoli_genome.tsv --output-prefix test_ecoli_genome --
amrrules --input tests/data/input/test_kleb_SGH10.tsv --output-prefix test_kleb_SGH10 --orga
```

How to share feedback

- Post an issue: https://github.com/AMRverse/AMRrules/issues
- Slack channel #codequeries
- Email esgem.amr@gmail.com

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R ☆ 15 ⑂ 9

AMRrulemakeR Public

What the Package Does (One Line, Title Case)

R ☆ 1

AMRrulebrowser Public

Web browser to interactively explore AMRrules

JavaScript ⑂ 1

AMRrulevalidator Public

Validates draft AMRrules files before importation into interpretation engine

Python

AMRrules

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AMRruleBrowser

amrverse.github.io/AMRrulebrowser

AMRrules **Browser**

Select an organism to browse: All organisms

Search in results: Enter search term...

Search | Clear

Browsing: all organisms

Displaying 709 row(s).

Download TSV | Columns to display

ruleID	organism	gene	nodeID	protein accession	HMM accession	mutation	variation type	gene context	drug	drug class	phenotype	clinical category	evidence code	evidence grade	evidence limitations
ACI0001	Acinetobacter baumannii	blaOXA-23_fam	blaOXA-23_fam	-	NF000266.2	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	moderate	lacks evidence of the degree to which MIC is affected
ACI0002	Acinetobacter baumannii	blaOXA-23	blaOXA-23	WP_001046004.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	high	-
ACI0003	Acinetobacter baumannii	blaOXA-27	blaOXA-27	WP_063862443.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	high	-
ACI0004	Acinetobacter baumannii	blaOXA-49	blaOXA-49	WP_063864111.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	moderate	lacks evidence of the degree to which MIC is affected
ACI0005	Acinetobacter baumannii	blaOXA-146	blaOXA-146	WP_063861044.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	moderate	lacks evidence of the degree to which MIC is affected
ACI0006	Acinetobacter baumannii	blaOXA-165	blaOXA-165	WP_063861123.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	moderate	lacks evidence for this allele
ACI0007	Acinetobacter baumannii	blaOXA-166	blaOXA-166	WP_063861129.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	moderate	lacks evidence for this allele
ACI0008	Acinetobacter baumannii	blaOXA-167	blaOXA-167	WP_063861133.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant	moderate	lacks evidence for this allele

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AMRruleBrowser

amrverse.github.io/AMRrulebrowser

AMRrules **Browser**

Select an organism to browse:

- ✓ All organisms
- Acinetobacter baumannii
- Enterobacter hormaechei
- Enterobacter roggenkampii
- Enterococcus faecalis
- Enterococcus faecium
- Escherichia coli
- Klebsiella grimontii
- Klebsiella huaxiensis
- Klebsiella michiganensis
- Klebsiella oxytoca
- Klebsiella pasteurii
- Klebsiella spallanzanii
- Klebsiella pneumoniae
- Neisseria gonorrhoeae
- Pseudomonas aeruginosa
- Salmonella enterica
- Staphylococcus aureus
- Yersinia enterocolitica
- Yersinia pseudotuberculosis

Search in results:

Enter search term...

Search | Clear

Download TSV | Columns to display

		gene	nodeID	protein accession	HMM accession	mutation	variation type	gene context	drug	drug class	phenotype	clinical category	evidence code	evidence grade	evidence limitations
		blaOXA-23_fam	blaOXA-23_fam	-	NF000266.2	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	moderate	lacks evidence of the degree to which MIC is affected
		blaOXA-23	blaOXA-23	WP_001046004.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	high	-
		blaOXA-27	blaOXA-27	WP_063862443.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	high	-
		blaOXA-49	blaOXA-49	WP_063864111.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	moderate	lacks evidence of the degree to which MIC is affected
ACI0005	Acinetobacter baumannii	blaOXA-146	blaOXA-146	WP_063861044.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	moderate	lacks evidence of the degree to which MIC is affected
ACI0006	Acinetobacter baumannii	blaOXA-165	blaOXA-165	WP_063861123.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	moderate	lacks evidence for this allele
ACI0007	Acinetobacter baumannii	blaOXA-166	blaOXA-166	WP_063861129.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant evidence	moderate	lacks evidence for this allele
ACI0008	Acinetobacter baumannii	blaOXA-167	blaOXA-167	WP_063861133.1	-	-	Gene presence detected	acquired	-	carbapenem	nonwildtype	R	ECO:0001103 natural variation mutant	moderate	lacks evidence for this allele

AMRrules

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AMRrules **Browser**

amrverse.github.io/AMRrulebrowser

Select an organism to browse: Neisseria gonorrhoeae

Search in results: Enter search term...

Search | Clear

Browsing: Neisseria gonorrhoeae

Download TSV | Columns to display

Displaying 124 row(s).

ruleID	organism	nodeID	protein accession	mutation	variation type	gene context	drug	drug class	phenotype	clinical catego
NGO0012	Neisseria gonorrhoeae	blaTEM-1	WP_000027057.1	-	Gene presence detected	acquired	benzylpenicillin	-	nonwildtype	R
NGO0013	Neisseria gonorrhoeae	blaTEM-135	WP_015058868.1	-	Gene presence detected	acquired	benzylpenicillin	-	nonwildtype	R
NGO0014	Neisseria gonorrhoeae	folP	WP_003702942.1	p.Arg228Ser	Protein variant detected	core	-	sulfonamide antibiotic	nonwildtype	R
NGO0015	Neisseria gonorrhoeae	folP	WP_003702942.1	p.Phe31Leu	Protein variant detected	core	-	sulfonamide antibiotic	nonwildtype	R
NGO0016	Neisseria gonorrhoeae	folP	WP_003702942.1	p.Gly194Cys	Protein variant detected	core	-	sulfonamide antibiotic	nonwildtype	R
NGO0017	Neisseria gonorrhoeae	folP	WP_003702942.1	p.194_195insSerGly	Protein variant detected	core	-	sulfonamide antibiotic	nonwildtype	R

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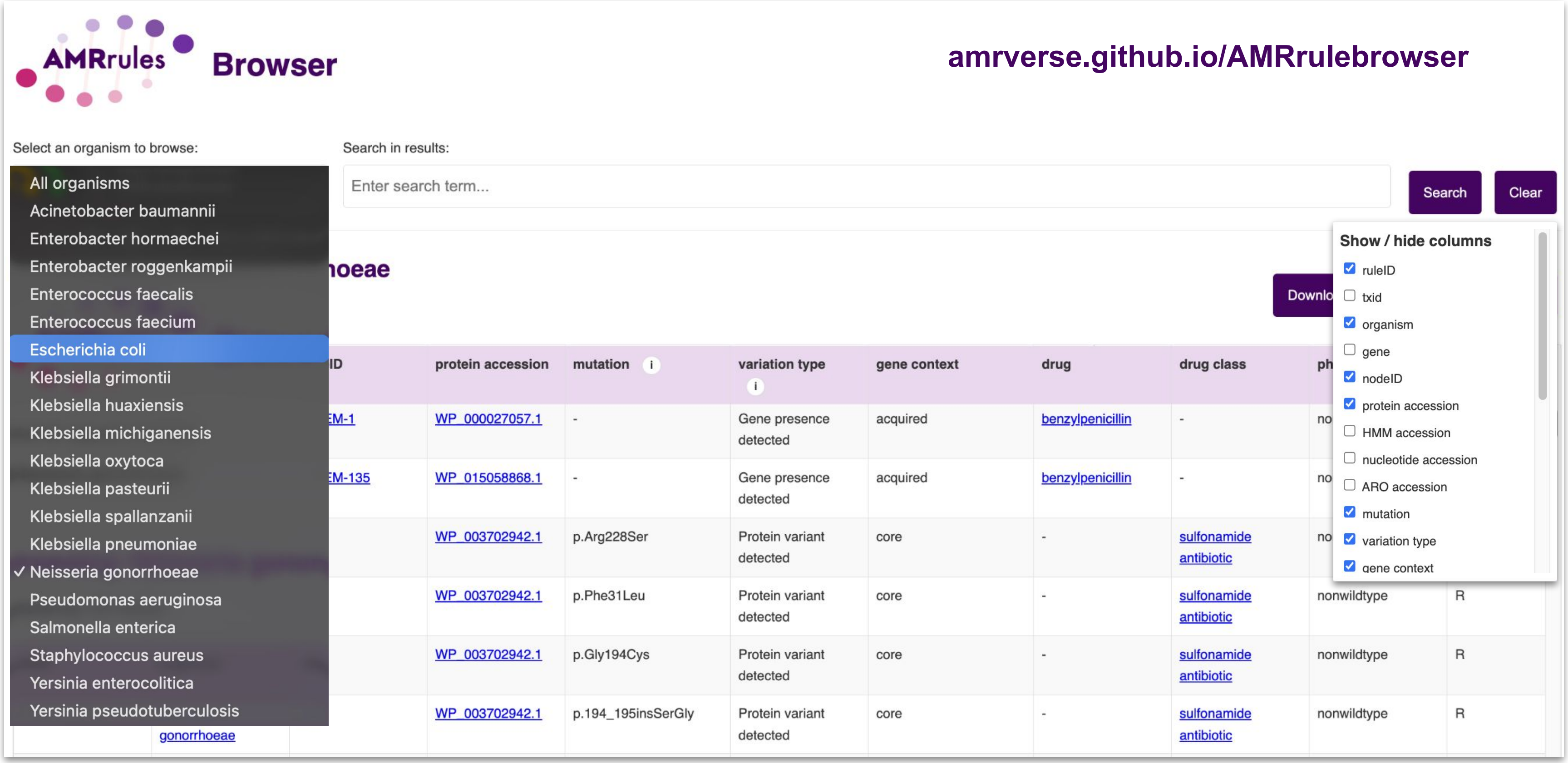

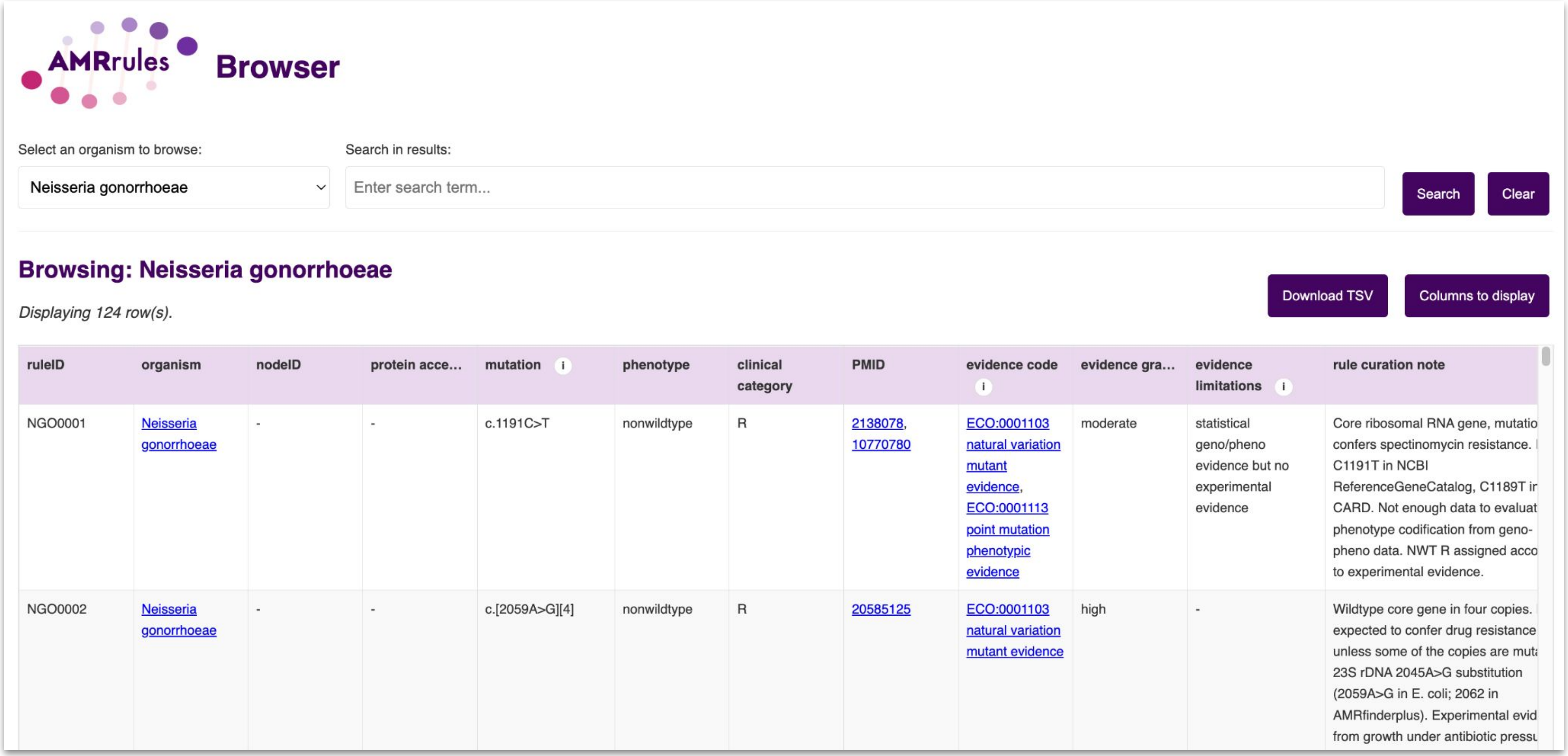
AMRrules Browser
Select an organism to browse:
Neisseria gonorrhoeae
Search in results:
Enter search term...
Search
Clear
Browsing: Neisseria gonorrhoeae
Displaying 124 row(s).
Download TSV
Columns to display
ruleID
organism
nodeID
protein acce...
mutation
phenotype
clinical category
PMID
evidence code
evidence gra...
evidence limitations
rule curation note
NGO0001
Neisseria gonorrhoeae
-
-
c.1191C>T
nonwildtype
R
2138078, 10770780
ECO:0001103 natural variation mutant evidence, ECO:0001113 point mutation phenotypic evidence
moderate
statistical geno/pheno evidence but no experimental evidence
Core ribosomal RNA gene, mutatio
confers spectinomycin resistance.
C1191T in NCBI
ReferenceGeneCatalog, C1189T ir
CARD. Not enough data to evaluat
phenotype codification from geno-
pheno data. NWT R assigned acco
to experimental evidence.
NGO0002
Neisseria gonorrhoeae
-
-
c.[2059A>G][4]
nonwildtype
R
20585125
ECO:0001103 natural variation mutant evidence
high
-
Wildtype core gene in four copies.
expected to confer drug resistance
unless some of the copies are mut
23S rDNA 2045A>G substitution
(2059A>G in E. coli; 2062 in
AMRfinderplus). Experimental evid
from growth under antibiotic pressu

Please provide feedback on the AMRruleBrowser

- Explore the browser: https://amrverse.github.io/AMRrulebrowser/
- Look examples of usage at README in AMRrulebrowser repo
    - https://github.com/AMRverse/AMRrulebrowser/blob/main/README.md

How to share feedback

- Post an issue: https://github.com/AMRverse/AMRrulebrowser/issues
- Slack channel #codequeries
- Email esgem.amr@gmail.com

Agenda

1. **AMRrules Python package - for March release**
   - Genome summary reports
   - Finalising core gene rules
2. **AMRrulebrowser**
3. **AMRgen R package**
   - New functions
4. **Planning**
   - Geno-pheno data-driven rules
   - Manuscript plans

https://github.com/AMRverse

AMRrules Public
Forked from katholt/AMRrules
Interpretative standards for AMR genotypes
Python ☆ 27 Forks 4

ESGEM-AMR Public
This is the webpage of the ESGEM-AMR Working Group
HTML ☆ 15 Forks 4

$\boxed{\textbf{AMRgen}}$ Public
The AMRgen package provides tools for interpreting antimicrobial resistance (AMR) genes, integrating genomic data with phenotypic antimicrobial susceptibility testing (AST) data, and calculating ge...
R ☆ 15 Forks 9

AMRrulemakeR Public
What the Package Does (One Line, Title Case)
R ☆ 1

AMRrulebrowser Public
Web browser to interactively explore AMRrules
JavaScript Forks 1

AMRrulevalidator Public
Validates draft AMRrules files before importation into interpretation engine
Python

AMRgen updates

- Phenotype import functions (instruments, EBI, NCBI)
- Genotype import functions (AMRfp versions, Abricate, …)
- Download functions (EBI geno/pheno, NCBI pheno)
- Export functions
- PPV
- Concordance function
- Interpretations against ECOFF recorded as NWT/WT (AMR pkg)

AMRgen updates

- Phenotype import functions (instruments, EBI, NCBI):

  `import_ast()` - import and process antimicrobial phenotype data from common sources

  `import_ebi_ast()` - import and process antimicrobial susceptibility phenotype data from the EBI AMR web portal

  `import_ebi_ast_ftp()` - import and process antimicrobial susceptibility phenotype data from the EBI AMR portal FTP site

  `import_microscan_ast()` - import and process antimicrobial phenotype data exported from MicroScan instruments

  `import_ncbi_ast()` - import and process antimicrobial susceptibility phenotype data from the NCBI AST browser

  `import_ncbi_biosample()` - import and process antimicrobial phenotype data retrieved from NCBI BioSamples

  `import_sensititre_ast()` - import and process antimicrobial phenotype data exported from Sensititre instruments

  `import_vitek_ast()` - import and process antimicrobial phenotype data exported from Vitek instruments

  `import_whonet_ast()` - import and process antimicrobial phenotype data from WHONET files

AMRgen updates

- Genotype import functions

`import_amrfp()` - import and process AMRFinderPlus results

`import_amrfp_ebi()` - import EBI-processed AMRFinderPlus genotypes

`import_amrfp_ebi_ftp()` - import EBI-processed AMRFinderPlus genotypes from FTP

`import_amrfp_ebi_web()` - import EBI-processed AMRFinderPlus genotypes from Web

Note: genotype results are processed and not all fields returned by AMRfinderplus are included.

AMRrules

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AMRgen updates

Note: genotype results are processed and not all fields returned by AMRfinderplus are included.

- Download functions (EBI geno/pheno, NCBI pheno)

  `download_ebi()` - download antimicrobial genotype or phenotype data from the EBI AMR portal

  `download_ncbi_ast()` - download NCBI antimicrobial susceptibility testing (AST) data

```
# Download AST data
ast <- download ncbi_ast("Klebsiella
quasipneumoniae")
ast <- download ebi(species =
"Staphylococcus aureus", reformat =
TRUE)

# Download Klebsiella quasipneumoniae
data, filter to amikacin and
ampicillin
ast <- download ncbi ast(
  "Klebsiella quasipneumoniae",
  antibiotic = c("amikacin", "Amp")
)

# Download and reformat for AMRgen
workflow with EUCAST interpretation
ast <- download ncbi ast(
  "Klebsiella quasipneumoniae",
  reformat = TRUE,
  interpret_eucast = TRUE)
```

AMRgen updates

- Export functions

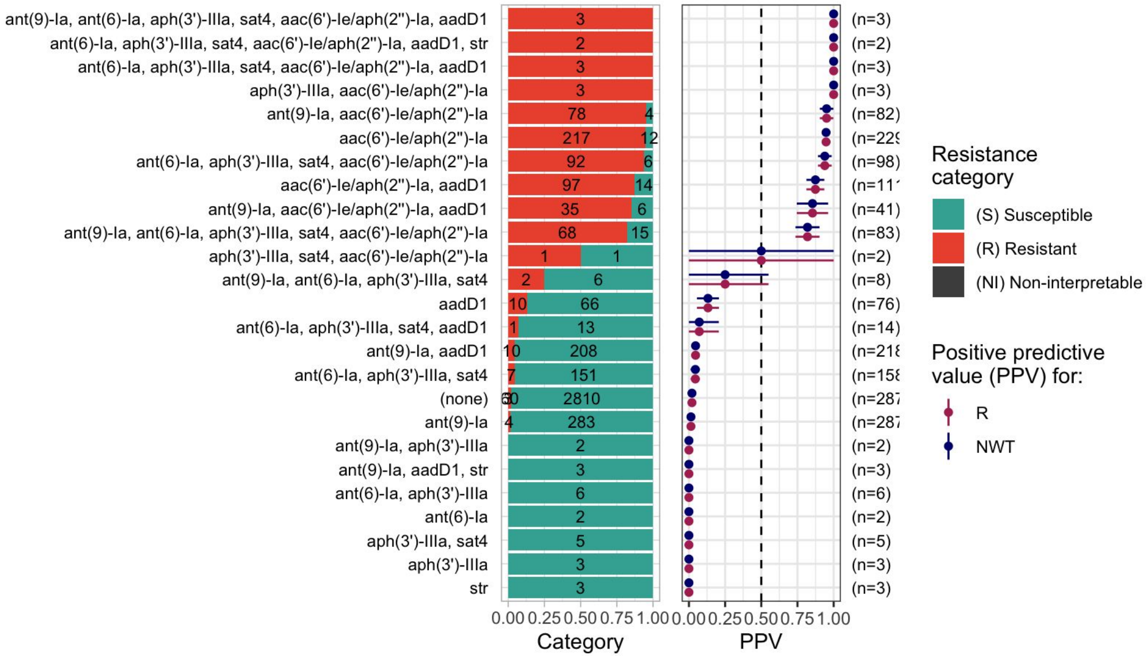
`export_ast()` - export AST data

`export_ebi_antibiogram()` - export EBI Antibiogram

`export_ncbi_biosample()` - export NCBI BioSample Antibiogram

AMRgen updates

- PPV function: `ppv()`



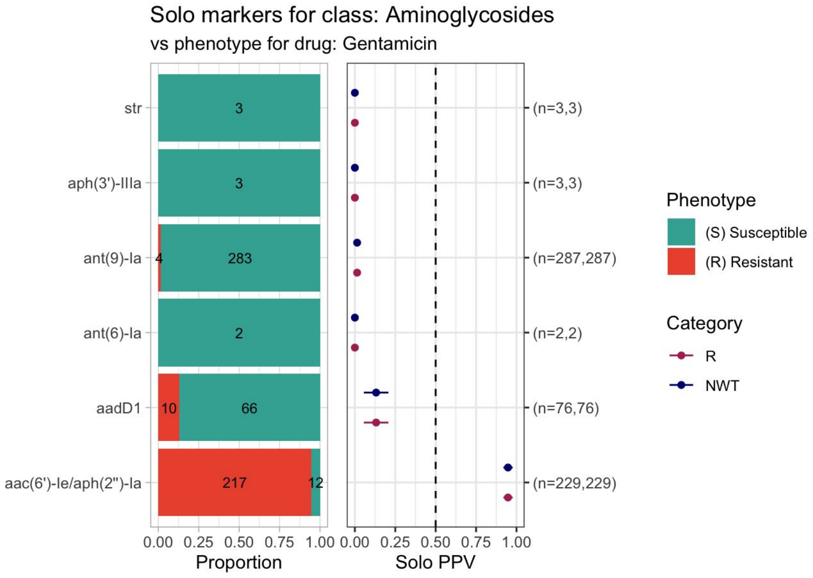
AMRrules

VME (Very Major Error)
Proportion of truly resistant isolates not predicted as such from genotype (target <1.5%).

ME (Major Error)
Proportion of truly susceptible isolates incorrectly predicted resistant from genotype (target <3%).

AMRgen updates

- Concordance function: `concordance()`




```
--- Outcome: R ---
Samples: 4322 | Markers: 1
Markers used: aac(6')-Ie/aph(2'')-Ia

Confusion Matrix:
          Truth
Prediction    1    0
         1  601   60
         0   94 3567

Metrics:
  Sensitivity : 0.8647
  Specificity : 0.9835
  PPV         : 0.9092
  NPV         : 0.9743
  Accuracy    : 0.9644
  Kappa       : 0.8653
  F-measure   : 0.8864
  VME         : 0.1353
  ME          : 0.0165
```

```
--- Outcome: NWT ---
Samples: 4322 | Markers: 1
Markers used: aac(6')-Ie/aph(2'')-Ia

Confusion Matrix:
          Truth
Prediction    1    0
         1  601   60
         0   94 3567

Metrics:
  Sensitivity : 0.8647
  Specificity : 0.9835
  PPV         : 0.9092
  NPV         : 0.9743
  Accuracy    : 0.9644
  Kappa       : 0.8653
  F-measure   : 0.8864
  VME         : 0.1353
  ME          : 0.0165
```

AMRrules

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AMRgen updates

- AMR pkg now records interpretations against ECOFF as NWT/WT

```
> head(ecoli_ast)
# A tibble: 6 × 17
  id           drug_agent      mic  disk pheno_clsi  ecoff guideline method pheno_provided spp_pheno
  <chr>        <ab>          <mic> <dsk> <sir>       <sir> <chr>     <chr>  <sir>          <mo>
1 SAMN36015110 CIP         <128.00    NA   NI          NI  CLSI      NA       R            B_ESCHR_COLI
2 SAMN11638310 CIP          256.00    NA   R          NWT  CLSI      NA       R            B_ESCHR_COLI
3 SAMN05729964 CIP           64.00    NA   R          NWT  CLSI      Etest    R            B_ESCHR_COLI
4 SAMN10620111 CIP          >=4.00    NA   R          NWT  CLSI      NA       R            B_ESCHR_COLI
5 SAMN10620168 CIP          >=4.00    NA   R          NWT  CLSI      NA       R            B_ESCHR_COLI
6 SAMN10620104 CIP         <=0.25     NA   S           NI  CLSI      NA       S            B_ESCHR_COLI
```

AMRgen next steps

- Import genotypes from other sources (Abricate ResFinder, CARD RGI, hAMRonization, etc.)
- Add unit tests
- Fix drug class/subclass categories
- Submit package to CRAN

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   - Manuscript plans

AMRrules v2: Data-driven development of rules for acquired resistance

- **Protocols in development for**
  - Using AMRrulemakeR package to propose rules from geno-pheno data
  - Manual expert review and curation of proposed rules
    It is important that all rules are manually reviewed and curated by experts who have knowledge of the organism, the drug, the resistance mechanisms, and relevant literature

- **AMRrulemakeR code to be finalised in next few weeks**
  - Data-rich groups (*E. coli, K. pneumoniae, S. enterica, N. gonorrhoeae, E. faecium, S. aureus*) testing
  - Other groups should work on sourcing data if possible (we can assist)
  - If insufficient geno-pheno data, focus on defining rules for all markers currently reported by AMRfp in Allthebacteria project based on literature

2026 Planning - Manuscripts

1. **AMRgen R package** (Q1)
2. **AMRrules** (Q2)
    - *Concept* - aims, principles, working group model
    - *Specification* (v1.0) - format for how rules are specified, use of standard ontologies, new/modified ontologies/syntax
    - *Implementation* - how rules are applied to interpret AMRfp output (Python package v1.0)
3. **Core gene rules** (Q2)
    - Why interpreting core gene rules is important but challenging
    - Approach to defining rules
    - Summary of rules - comparison across pathogens, gaps in knowledge
4. **Quantitative approach and acquired gene rules** (Q4)

2026 Planning

1. **Fixed monthly meeting dates** - *Tuesdays 9am/5pm UK time*
   - Mar 17, Apr 28, May 19, Jun 23, Jul 21, Aug 18
2. **Wellcome AMR Conference** (23-25 March, Hinxton, UK)
   - ESGEM-AMR Catch-up during first tea break, Monday 15:30
3. **ESCMID Global** (17-21 April, München, Germany)
   - ESGEM Business meeting, Saturday 18 April, 11-12h Room ICM 2
4. **Collaboration with WHO AMR team on AMR Gene Catalog**

Questions? / Any other business?

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AMRrules

amrrules.org